# BHANU RAHUL PEDAMALLA

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**Software Engineer, Data Pipelines and Full-Stack | M.S. Data Science**

## SUMMARY

Software engineer, M.S. Data Science, shipping production systems end to end as the only engineer. Built a cancer casefinding pipeline over HL7 v2 and FHIR feeds with 278 of 279 case records correct, a shelf-scan conflation engine at 97.9% with zero wrong records, and a robot telemetry pipeline proven lossless across 63 trips. Designed and shipped X Reacts alone: a live, mobile-first creator platform with real-time chat, real payments, creator and admin portals, and a video-sync engine. Staged, versioned pipelines; every stage scored before the next; rules first, a language model only where rules stop; every number reproducible from one run.

## TECHNICAL SKILLS

**Data pipelines:** Python, pandas, PyArrow/Parquet, DuckDB, SQL. HL7 v2 (ORU, ADT), FHIR R4, NAACCR, ICD-O-3, OMOP CDM. Entity resolution and conflation, golden records with survivorship rules, schema and data versioning, lineage, deterministic reruns, manifests and check lines, quarantine and dedupe, scorers against ground truth.

**Applied AI:** Claude API (Haiku, Sonnet), prompt caching, measured model and threshold selection, confidence gating, cost per answer. rapidfuzz, regex and rules-based normalization.

**Front end:** React, Vite, JavaScript, HTML5, CSS3, Tailwind CSS, responsive mobile-first design down to 375 px, reusable components, real-time UI, Playwright for page checks.

**Back end and cloud:** Node, Firebase (Auth, Firestore, Cloud Functions, Storage, Hosting), REST APIs, payment integration with server-side verification, security rules and role-based access, Cloudflare, Git and GitHub.

## PROJECTS

### Cancer Casefinding Pipeline

Oct 2026

*Live:* registry-replay.web.app *Report:* registry-replay.web.app/pipeline *Code:* GitHub: casefinding-pipeline

*Problem: a hospital cancer registry receives pathology, admissions, clinical notes and billing from several systems, out of order and with mismatched identities. Output: one trusted record per cancer case, coded to NAACCR, exported for analytics.*

- Simulated three hospitals sending HL7 v2.5.1 ORU and ADT (A04, A08, A40), FHIR R4 bundles and billing CSV: 1,320 Synthea patients, 3,554 arrivals, 279 reportable cases, 287 look-alikes, 19 planted problem types, and an ICD-O-3 coding change on day 11.
- Built the pipeline: quarantine with a reason per rejected message; parse and normalize; patient resolution on SSN, merge links and a demographic threshold measured clean across 5 to 10; reconciliation of documents into cases; a reportability ladder of SEER rules with a language model only for the 61 reports no rule could decide; record versioning with a test that the record survives the coding change; NAACCR flat and OMOP exports served from DuckDB.
- Result: 279 records, 278 right, 0 missed, 0 duplicate cases, 0 contested; every one of 3,554 arrivals accounted for as accepted, duplicate, quarantined or non-reportable. Every planted problem caught. LLM bill $0.0054, prompts cached, rerun byte-identical in 109 s.
- Built the public replay page and report page: the feed plays on load, tapping a record pins its card, every number is baked at deploy and recounted at load, checked at 375 px and 1200 px with zero console errors.

### Shelf-Scan Conflation Engine

Oct 2026

*Live:* shelf-replay.web.app/web *Report:* shelf-replay.web.app/pipeline *Code:* GitHub: shelf-conflation-engine

*Problem: a badge camera worn by store staff reads shelf labels as they walk. Reads arrive damaged, duplicated and re-sent, and the product catalog changes under them. Output: one record per shelf spot, matched to the catalog and served against the planogram.*

- Simulated the feed with real data: 3 stores, 7 aisles, 300 real products from Open Food Facts, 12 badge walks over 3 days, restocking between days, doubles in a frame, a second badge, re-sent frames, about 1 in 100 rows damaged, and a catalog change (3 renames, 2 new sizes with new barcodes, 1 discontinued, 1 new).
- Built a 9-stage pipeline: ingest with quarantine and a reason for every dropped row; normalize; barcode and exact-text match; close spelling with the threshold read off a measured curve; same-spot and other-day evidence; a language model only for the 159 spots nothing else could name, accepted at "sure" only.
- Golden record per shelf spot with survivorship by method rank (barcode > exact > close > same spot > other day > model), lineage across the catalog change, a test that the record survives the change, and DuckDB serving against the planogram with a compliance table per store, aisle and shelf.
- Result: 5,927 lines in, 5,705 usable reads, 5,586 named right (97.9%), 3 wrong and none reached the record; 1,763 records, 0 wrong, 1 contested, 18 open; compliance 96.0% against a 96.5% truth, gap reconciled exactly.
- Measured before choosing: Haiku vs Sonnet on 696 spots (both 0 wrong at "sure", Haiku 13.6x cheaper), four survivorship rules against the truth, slots vs clusters for spot grouping. LLM bill for the whole feed: $0.02.
- Every stage scored before the next, byte-identical reruns, full run in 20 s with no API key. The report page counts every number from the data files at load.

### Robot Telemetry Pipeline

Aug 2026

*Live:* rovi-replay.web.app/sim *Report:* rovi-replay.web.app/pipeline *Code:* GitHub: rovi-pipeline

*Problem: a hospital transport robot produces about 15,000 messages per trip across 13 sensor streams, and hospital WiFi drops in lifts. Every message must reach the cloud once, in order, provably.*

- Built a 7-stage pipeline: record every message to the robot's own disk before any transmission; transport; validate with the message id as primary key so duplicates are harmless; cold storage; refined table ordered by event time rather than arrival time. The robot keeps its copy until the server confirms the checksum, so a damaged segment can be repaired.
- Simulated 63 trips with 38 network blackouts, 11 duplicate deliveries and 2 files corrupted in transit. Result: 915,068 messages produced, 915,068 in cold storage, 915,068 in the refined table, none lost. Every message ends as accepted, duplicate or quarantined with a reason, and the counts sum to what arrived.
- A prove script plants seven faults and checks each is caught; a page check confirms the write-up has no hardcoded numbers.

### Penn Station Transfer-Risk Forecasting

2025

*Graduate project, University of Texas at Arlington*

- Aggregated 142,844 multimodal records and 82,959 real-time arrivals from multiple sources into 299,186 engineered scenarios.
- Cross-validated a predictive model and recommended a 2-3 minute transfer buffer that cuts failure risk from 75% to under 20%, affecting 600,000+ daily riders.

## EXPERIENCE

### Founder & Engineer, X Reacts, Gainesville, FL

Mar 2025 - Present

*Live platform where creators react to videos on stream and viewers pay to take part.*

- Designed and shipped the whole product alone: React/Vite front end, Firebase back end, 19+ surfaces (onboarding, creator studio, viewer page, earnings and withdrawals, admin portal, real-time chat) and a mobile admin app.
- Wrote the video-sync algorithm that holds the viewer's player 4.6 s behind the creator's live stream through pauses, seeks and resumes, using playback-rate correction instead of hard seeks.
- Built the checkout, payment and virtual-currency flow on real transactions with automated splits and refunds, and client-side NSFW moderation on uploads.
- Ran a security hardening pass that moved trust from the client to the server, locked database rules and closed every identified vulnerability.
- Owns the data layer, creator onboarding, support and payouts under Pedamalla LLC.

### Founder & Director, Creator Management Agency, India

2022 - Dec 2023

- Ran a cross-functional team across editing, design, scripting and account management for content creators; used performance data to set priorities and grow client reach.

## EDUCATION

### Master of Science, Data Science, University of Texas at Arlington

Jan 2024 - Dec 2025

*GPA 3.9 / 4.0, STEM-designated*

### B.Tech., Computer Science & Engineering, Chalapathi Institute of Technology, India

Jun 2018 - Jul 2022

## ADDITIONAL

**Work authorization:** Authorized to work in the United States.